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W5 Semantic Abstraction

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READINGS Pylyshyn, *Computation and Cognition* (1984), chs. 1–2; Newell, "The Knowledge Level" (1982), excerpts

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1 Two kinds of abstraction

- **Abstraction in scientific explanation** vs **abstraction in cognition.**
  - Are these independent? Perhaps not...
- **Representation as abstraction?** Does representation always involve a kind of abstraction?
- **Semantic abstraction.** Abstraction in cognition is part of what drives specific kinds of abstraction in science, namely semantic abstraction.

2 The aerial view

- **A cognitive level of explanation.** Pylyshyn wants to convince us of the importance of a cognitive or semantic level of explanation, characterized by a distinctive cognitive vocabulary.
- **Over and above.** Pylyshyn wants to argue for the importance of this over and above a purely neurophysiological level of explanation, or a "functional" or "algorithmic" level of explanation.

3 Chapter 1: The vocabulary of cognition

The 911 case

- **The case.** "A pedestrian is walking along a sidewalk. Suddenly the pedestrian turns and starts to cross the street. At the same time, a car is traveling rapidly down the street toward the pedestrian. The driver of the car applies the brakes. The car skids and swerves over to the side of the road, hitting a pole. The pedestrian hesitates, then goes over and looks inside the car on the driver's side. He runs to a telephone booth at the corner and dials the numbers 9 and 1." (p. 3)
- **What do they do next?** And what explains their next action?
- **Two possible explanations.**
  - **A neurophysiological explanation.** "by innervating a cycle of opposing tenser [sic] muscles, which causes the position of one foot to be placed ahead of the other" (p. 4)
  - **A cognitive-level explanation.** "the pedestrian *perceived* the collision, *recognized* it as an event that is classified as an accident, *inferred* that there might be an injury, . . . *recalled* the number for emergencies, and dialed the number with the *intention* of seeking help." (p. 5)
- **One is better.** There's a sense that one explanation is better than the other.
  - "this class of vocabulary—as opposed to a physical or biological class—is precisely what appears to be needed actually to explain the *why*, *what*, and *how* of the event." (p. 5)
  - Why?

Explanatory alternatives

- **Contrast class.** All explanations come with a contrast class, or a space of possible variation in the behavior being explained.
  - "Any time we seek to explain a phenomenon we always do so in relation to some way of conceptualizing, or dividing, the space of possible alternatives" (p. 16)
- **Describing the situation.** The contrast class will vary depending on how you describe the situation in question.
  - **Cognitive alternatives.** What if their *knowledge* differed? "if the pedestrian in our example does not perceive the event as an emergency, does not know what to

do, or does not remember the phone number, the entire sequence could be quite different." (p. 6)

  - **Neurophysiological alternatives.** A space of alternatives differentiated by things that could have gone differently at the neurophysiological level.

Cognitive vocabulary

- **Vocabulary.** The behavior being explained has to be specified in a certain vocabulary, which implicitly sets the contrast class.
- **Cognitive vocabulary.** "talk of 'knowing,' 'perceiving as,' 'being informed of,' and so on" (p. 6); "goals and beliefs" (p. 7)
  - **How we see things.** A key feature of the cognitive vocabulary is that it captures how we characterize things mentally, rather than how they are.
  - "And, of course, it is what we see things as that determines their effect on our behavior. When we see a certain red light *as a traffic light*, we stop. . . That, in a nutshell, is why the physical properties of stimuli do not determine our behavior." (p. 15)
- **Explanatory success.** Only the cognitive story (not the neurophysiological story) explains why the *action of pressing '1'* occurred at the relevant level of grain.
- **A better argument.** This is better than an argument you sometimes hear: because the thing to be explained is described in cognitive terms, no physiological thing can derive it. (This is where bridge laws come in.)

Capturing generalizations

- **A worry.** Does selecting the behavior force the explanation? Why was that the right way to describe behavior in the first place?
- **Generalizations.** For Pylyshyn, certain vocabulary will capture generalizations that other vocabulary can't.
- **A whole range of cases.** An explanation couched in the cognitive vocabulary would explain a whole range of cases: different kinds of action, different kinds of evidential source.
  - "regardless of how a person learns the emergency number, regardless of how he or she comes to perceive a situation as an emergency, and regardless of how the person's limbs are moved in dialing the number, a single, general principle is implicit" (p. 11)

- **Levels and generalizations.** The lower level can capture variations that can't be captured at the top level, but the top level can capture generalizations that can't be captured below.
  - "Physical and neurophysiological terms ... distinguish aspects of the world and of behavior that are equivalent with respect to their psychological import, and sometimes fail to make distinctions that are psychologically relevant." (p. 17)

Questions

- **Level as language.** Pylyshyn is thinking of the cognitive vocabulary in a classical theory sense, where each science could be articulated as a set of sentences in a predicate calculus, and have a literal vocabulary.
- **Are levels really representational?** Are levels really so representational that we should think of each as associated with different representational systems? Or is vocabulary more metaphorical, a set of rules or principles?
- **What is the key function of "vocabulary"?** Is it something specific about the kind of categories we're dealing with? or the domain of properties we're drawing from?

4 Chapter 2: The explanatory role of representations

The big picture

- **Three levels.** Like Marr, Pylyshyn argues for a three-level description of cognition: a neurophysiological level, a symbolic level, and a semantic level.
- **The problem.** Why is the semantic level necessary, given a symbolic level?

The symbol level: representations and functional architecture

- **The symbol level** = the functional level = the algorithmic level.
- **Representations.** The representations are what vary from state to state.
- **Functional architecture.** The functional architecture is relatively fixed, and helps guide the use of these representations. Functional architecture is what makes computation possible.
  - "Two people whose beliefs and goals are all the same may nevertheless be functionally distinguishable if they have, say, slightly different memory-retrieval mechanisms, short-term memory capacities, degrees of alertness" (p. 31)

- **Pylyshyn's razor.** Later, Pylyshyn offers a famous principle: you ought not attribute representations when you can explain the same behavior through functional architecture.

Brentano's problem

- **Contents.** The contents of your thoughts are not something you're necessarily causally connected to. They describe things that are false or do not exist.
  - "The case is even clearer when we find people's behavior being a function of such nonexistent things as Santa Claus, the pot of gold at the end of the rainbow, a hoped-for raise, or beliefs that are false" (p. 25)
- **As materialists.** As materialists we should be suspicious of attempts to add to our scientific vocabulary entities which don't seem to play an obvious causal role. This is essentially Brentano's problem.

Pylyshyn's resolution

- **A physical code or a symbol.** "The answer I ultimately give to this puzzle is that the causes of the behavior are not literally numbers, anticipated future events, or other 'intentional objects' but rather some physically instantiated internal representation of such things, that is, a physical code or a symbol." (p. 26)
- **Equivalence classes.** "the codes or symbols are equivalence classes of physical properties which, on one hand, cause the behavior to unfold as it does, and on the other, are the bearers of semantic interpretations that provide the needed higher-level principle for their individuation and for stating the generalizations." (pp. 39–40)

My take

- **Reifying contents.** We've fallen into a style of thinking: reifying contents, thinking of them as objects, and wondering how they could cause anything.
- **Contents as high-level properties.** A better way to think of content is as a property or description at a higher level of abstraction.
- **Causal profiles.** Contents themselves don't cause anything. But states having contents do have distinctive causal profiles. As scientists we use the contents to characterize a certain class of states.

The explanation principle

- **Ontological conservativeness.** We should only posit contents when our explanatory needs require us to. If we can explain the same phenomena at a lower level, we ought to do that instead.
  - "the notion of representation is necessary only in the context of explanation; it is needed to state generalizations concerning the behavior of systems under certain descriptions." (p. 26)
- **Explanatory methodology.** This explanation-first approach contrasts with a phenomenology-first or intuition-first approach. It is foundational to the cognitive scientific way of studying the mind.

Reasons for thinking of something as a representation (End Note 1)

- **(a) More than one way of representing.** "there will, in general, be more than one distinguishable way of representing something" (p. 46)
- **(b) Contrary to fact.** "the system can, in general, represent states of affairs that are contrary to fact." (p. 46)
- **(c) Not a direct causal consequence.** "there may also be an unlimited number of ways for the system to come to be in state *I* other than as a direct causal consequence of the occurrence of the corresponding state of the environment identified as *S*." (p. 46)
- **(d) Regularities in the domain.** "the state-to-state transition regularities that do occur in certain circumstances can be distinguished from other possible transitions that do not occur, by viewing the individual states *I* as representing some *S* and by appealing to regularities that exist in the domain of *S*" (p. 46)

Justifying the semantic level

- **Just the exercise of abstraction.** The attribution of a semantic level is just the exercise of abstraction, carried out just as it was to get to the algorithmic level (and, we might point out, to the biological level).
  - "The principle that leads us to postulate representational states (individuated by their content) that are distinct from functional states is exactly the same as the principle that leads us to postulate functional states that are distinct from physical states." (p. 32)
- **Something special.** There is something special about this abstraction, calling back to Brentano's problem: it takes into account the environment and environmental history of the organism.

  - **Non-semantic abstraction.** "These accounts make no reference to entities or properties other than those that determine what is sometimes referred to as the *location* of the system in its state or phase space. Because these are deterministic systems (what Ashby [1956] calls 'absolute' or 'state determined' systems), closed to the outside except for effects explicitly recognized as inputs or outputs, such an explanatory account is also closed and complete." (p. 25)
  - **Semantic abstraction.** "Accounts (4) and (5) differ in an important respect: Both make substantive reference to entities or properties that are not an intrinsic part of their state description, for example, *numbers* or *need for assistance*." (p. 25)
  - **Semantic abstraction II.** "It is important to recognize that such terms do not refer to properties of the system whose behavior we are explaining (that is, the computer or the person doing the phoning) nor necessarily to properties causally connected to such states or to the system's behavior, that is, the behavior of the system cannot be explained in terms of a chain of causal laws connecting such extrinsic entities as those mentioned in accounts (4) and (5) to the current state of the system." (p. 25)

- **Upward and outward.** The other forms of abstraction are in some sense system-internal: upward abstraction. This kind of abstraction is system-external as well: outward abstraction.

Terminology: internalism and externalism

- **Externalism.** Roughly, the idea that the content of one's mental states is determined in part by factors outside of the organism.
- **Internalism.** Roughly, the idea that the content of one's mental states is determined by factors internal to the organism.

Rationality

- **Mark of the semantic.** For Pylyshyn, a norm or constraint of rationality is a clear mark of the semantic level.
- **A background principle.** He gives a variety of examples in scientific explanation where rationality of some kind is presupposed.
  - "we do not feel compelled to add something like, 'and if the person was acting rationally at the time'; it is assumed to be part of the 'normal background conditions.' " (p. 34)

- **Newell.** Newell said something very similar. Pylyshyn says his semantic level and Newell's knowledge level are the same.
  - "The latter term is taken from Newell (1980, 1982), who uses the term 'knowledge level' where I use 'semantic level.' " (p. 32)

For discussion

- **Levels and Marr.** Is the semantic level the same as Marr's computational level?
  - **My take.** Pylyshyn is a bit unclear about non-conceptual, non-inferential computation. But interpreting him broadly, his semantic level and Marr's computational level are quite similar in spirit.
- **Levels and Newell.** Is Newell's knowledge level the same as Pylyshyn's semantic level?
  - **My take.** Newell's knowledge level describes a subset of semantic-level explanations, for particular systems that support a knowledge level. A perceptual system isn't necessarily best described in terms of a knowledge level.

5 Newell: the knowledge level

Computer system levels

- **What is a level?** "A level consists of a medium that is to be processed, components that provide primitive processing, laws of composition that permit components to be assembled into systems, and laws of behavior that determine how system behavior depends on the component behavior and the structure of the system." (p. 95)
- **Autonomy.** "Each level is defined in two ways. First, it can be defined autonomously, without reference to any other level." (p. 95)
- **Reduction.** "Second, each level can be reduced to the level below." (p. 95)
- **Variety.** "However, it is noteworthy how radically the levels differ. The medium changes from electrons and magnetic domains at the device level, to current and voltage at the circuit level, to bits at the logic level (either single bits at the logic circuit level or bit vectors at the register-transfer level), to symbolic expressions at the symbol level, to amounts of data (measured in data bits) at the configuration level." (pp. 95–96)
- **Basic features.** "Despite this variety, all levels share some common features." (p. 96)

  - **Point 1.** "Specification of a system at a level always determines completely a definite behavior for the system at that level (given initial and boundary conditions)." (p. 96)
  - **Point 2.** "The behavior of the total system results from the local effects of each component of the system processing the medium at its inputs to produce its outputs." (p. 96)
  - **Point 3.** "The immense variety of behavior is obtained by system structure, i.e., by the variety of ways of assembling a small number of component types (though perhaps a large number of instances of each type)." (p. 96)
  - **Point 4.** "The medium is realized by state-like properties of matter, which remain passive until changed by the components." (p. 97)

- **Realism.** "To sum up, computer system levels are a reflection of the nature of the physical world. They are not just a point of view that exists solely in the eye of the beholder. This reality comes from computer system levels being genuine specializations, rather than being just abstractions that can be applied uniformly." (p. 98)

The knowledge level

- **The proposal.** "I now propose that there does exist yet another system level, which I will call the knowledge level. It is a true systems level, in the sense we have just reviewed." (p. 98)

- **The agent.** "The system at the knowledge level is the agent. The components at the knowledge level are goals, actions, and bodies. Thus, an agent is composed of a set of actions, a set of goals and a body. The medium at the knowledge level is knowledge (as might be suspected). Thus, the agent processes its knowledge to determine the actions to take. Finally, the behavior law is the principle of rationality: Actions are selected to attain the agent's goals." (p. 98)

- **Above the symbolic level.** "The knowledge level sits in the hierarchy of systems levels immediately above the symbol level, as Fig. 3 shows. Its components (actions, goals, body) and its medium (knowledge) can be defined in terms of systems at the symbol level, just as any level can be defined by systems at the level one below." (pp. 98–99)

- **The Knowledge Level Hypothesis.** "There exists a distinct computer systems level, lying immediately above the symbol level, which is characterized by knowledge as the medium and the principle of rationality as the law of behavior." (p. 99)

- **Knowledge and rationality.** "Knowledge is intimately linked with rationality. Systems of which rationality can be posited can be said to have knowledge. It is unclear in what sense other systems can be said to have knowledge." (p. 100)

The structure of the knowledge level

- **The body.** "First, the agent has some physical body with which it can act in the environment (and be acted upon). We talk about the body as if it consists of a set of actions, but that is only for simplicity. It can be an arbitrary physical system with arbitrary modes of interaction with its environment." (pp. 100–101)
- **Knowledge.** "Second, the agent has a body of knowledge. This body is like a memory. Whatever the agent knows at some time, it continues to know." (p. 101)
- **Goals.** "Third, and finally, the agent has a set of goals. A goal is a body of knowledge of a state of affairs in the environment. Goals are structurally distinguished from the main body of knowledge. This permits them to enter into the behavior of the agent in a distinct way, namely, that which the organism strives to realize. But, except for this distinction, goal components are structurally identical to bodies of knowledge." (p. 101)
- **What behavior depends on.** "The behavior is to depend only what the agent knows, what it wants and what means it has for interacting physically with the environment." (p. 101)
- **Absence of structure.**
  - **No structural constraints.** "There are no structural constraints to the knowledge in a body, either in capacity (i.e., the amount of knowledge) or in how the knowledge is held in the body. Indeed, there is no notion of how knowledge is held (encoding is a notion at the symbol level, not knowledge level)." (p. 101)
  - **Essentially structure-less.** "This complete absence of significant structure in the agent is the first surprise, running counter to the common feature at all levels that variety of behavior is realized by variety of system structure (Section 3.1, Point 3). This is not fortuitous, but is an essential feature of the knowledge level. The focus for determining the behavior of the system rests with the knowledge, i.e., with the content of what is known. The internal structure of the system is defined exactly so that nothing need be known about it to predict the agent's behavior." (p. 101)
  - **Compare: possible-world analysis of propositions.**

The principle of rationality

- **Principle of rationality.** "If an agent has knowledge that one of its actions will lead to one of its goals, then the agent will select that action." (p. 102)
- **No mechanism.** "This principle asserts a connection between knowledge and goals, on the one hand, and the selection of actions on the other, without specification of any mechanism through which this connection is made. It connects all the components of the agent together directly." (p. 102)
- **Not sufficient to determine behavior.** "However, the principle is not sufficient to determine the behavior in many situations. Some of these situations can be covered by adding auxiliary principles. Thus, we can think of an extended principle of rationality, building out from the central or main principle, given above." (p. 102)
- **The lady or the tiger.** "Our knowledge-level model of the princess, even if it were to include her complete knowledge, would not tell us which she chose. But the failure of determinancy is not the model's." (p. 104)
  - **Radical incompleteness.** "Yet, radical incompleteness characterizes the knowledge level." (pp. 104–105)

The nature of knowledge

- **Knowledge defined.** "Whatever can be ascribed to an agent, such that its behavior can be computed according to the principle of rationality." (p. 105)
  - **Knowledge as...**
    * Knowledge
    * Belief
    * Commital mental states
    * Commital dispositions.
- **Relation to symbols and processes.** "This still leaves open the requirement for a physical structure for knowledge that can fill the functional role. In fact, that key role is never filled directly. Instead, it is filled only indirectly and approximately by symbol systems at the next lower level. These are total systems, not just symbol structures. Thus, knowledge, though a medium, is embodied in no medium-like passive physical structure." (p. 105)
  - **No physical structure.** "This failure to have the medium at the knowledge level be a state-like physical structure is the fourth surprise, running counter to the common feature at all levels of a passive medium (Section 3.1, Point 4). Again, it is an essential feature of the knowledge level, giving knowledge its special abstract and competence-like character." (p. 105)

Why is knowledge irreducible to symbols?

- **The question.** "Even granting the scheme of Fig. 4 to be possible, why cannot there simply be some physical structure that corresponds to knowledge?" (p. 107)
- **Knowledge is not finite.** "The answer in a nutshell is that knowledge of the world cannot be captured in a finite structure. The world is too rich and agents have too great a capability for responding. Knowledge is about the world. Insofar as an agent can select actions based on some truth about the world, any candidate structure for knowledge must contain that truth. Thus knowledge as a structure must contain at least as much variety as the set of all truths (i.e. propositions) that the agent can respond to." (p. 107)
- **Generative ability.** "The situation here is not really strange. The underlying phenomenon is the generative ability of computational systems, which involves an active process working on an initially given data structure. Knowledge is the posited extensive form of all that can be obtained potentially from the process." (p. 108)